

Thomas Thuan F. Le Nguyen

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Research Interests

Data Science, Machine Learning, Computer Vision, Medical AI, Agentic AI, and Modeling and Simulation.

Education

Florida International University

M.S. in Data Science and Artificial Intelligence

GPA: 3.92

Miami, FL

Expected Spring 2027

California State University San Marcos

B.S. in Computer Science, Minor in Mathematics

GPA: 3.34

San Marcos, CA

Fall 2023

Research Experience

Software Developer — MOSAIC (Mobile Sensing and Analytics) Lab

Oct 2025–Present

Florida International University

Miami, FL

Lab Director: Dr. Christian Poellabauer

Faculty Collaborator: Dr. Shanna Burke

- Design, develop, and test a mobile application for collecting participant-reported data on sleep, treatment, and daily well-being as part of a cancer survivorship research project funded in part by the Florida Cancer Innovation Fund.
- Develop and maintain a web-based administration platform for managing research studies and creating and configuring surveys and activities.
- Design and maintain database infrastructure supporting research study configuration, survey and activity management, and participant data management.
- Implement and maintain software functionality based on requirements provided by the research team and collaborators.
- Software developed is also used by Baptist Health South Florida in clinical surgery-related studies.

Undergraduate Researcher — Summer Scholars Program

Jun 2021–Aug 2021

California State University San Marcos

San Marcos, CA

Research Supervisor: Dr. Ali Ahmadinia

- Conducted Genentech-sponsored undergraduate research focused on computer vision for low-cost embedded systems.
- Developed and tested YOLOv4 TFLite object detection models using Google Colab.
- Evaluated models on a Coral Development Board, benchmarking inference speed (FPS) and mean Average Precision (mAP).
- Presented research findings at the Summer Scholars Undergraduate Research Program Final Research Showcase.

Presentations

Le Nguyen, Thomas Thuan F. “Object Detection Using Low Cost Embedded Systems”

Summer Scholars Undergraduate Research Program Final Research Showcase, California State University San Marcos, 2021. [\[Poster\]](#)

Selected Data Science & AI Projects

Electric Vehicle Ownership Prediction

- Analyzed and visualized electric vehicle adoption across Washington State counties using geospatial and demographic data.
- Examined relationships between EV adoption and factors including education, income, and political affiliation.
- Explored adoption trends over time and compared patterns across geographic regions.
- Developed visualizations to communicate findings and highlight regional trends.

Perceptron and Adaline Models

- Implemented and evaluated Perceptron and Adaline neural networks from scratch for vowel classification.
- Created a custom training set and three independent test sets to evaluate model performance.
- Compared model accuracy using quantitative results, graphs, and tables.

Payphone — AI-Enabled Point-of-Sale System

- Led a team of eight in developing a point-of-sale system incorporating artificial intelligence and machine learning for product identification and automated subtotal calculation.
- Integrated a database for inventory management across multiple store locations.

Teaching Experience

Instructional Student Assistant

Jan 2022–Dec 2023

CSIS Department, California State University San Marcos

San Marcos, CA

- Supported more than 500 students across courses in introductory programming, data structures, assembly language, digital circuits, and operating systems.
- Clarified advanced computer science concepts, graded assignments and examinations, and collaborated with course instructors.
- Communicated recurring student questions and challenges to instructors to improve course support.

STEM Instructor

May 2024–Aug 2024

CodeREV Kids

Los Angeles, CA

- Taught more than 300 students in computer programming, engineering, and 3D modeling.
- Conducted school enrichment sessions focused on circuitry and coding.

Lead Coding Instructor

Apr 2019–May 2024

Code Ninjas

Encinitas, CA

- Taught programming to more than 2,000 students using JavaScript, Lua, C#, and Java.
- Led multi-day coding camps and managed classes of 20–30 students.
- Conducted school enrichment sessions covering robotics, drones, and computer programming.

FLL Jr. Mentor

Aug 2017–Jun 2018

La Clase Magica

Solana Beach, CA

- Mentored kids from underserved communities for the FIRST Lego League Jr. competition.
- Helped plan and guided students to build a water system out of Legos.
- Taught the importance of clean water and illustrated the different water systems and cleaning services.

Industry Experience

Software Engineer Intern

Jun 2023–Aug 2023

Hunter Industries

San Marcos, CA

- Developed a minimum viable product for visualizing precise customer information.
- Developed methods for retrieving previously inaccessible data from customer controllers.
- Applied Test Driven Development (TDD), Agile, and Scrum methodologies.

Disney College Program — Merchandise

Aug 2024–May 2025

Walt Disney World

Orlando, FL

- Operated point-of-sale systems and assisted guests with purchases.
- Managed inventory, restocked merchandise, and maintained organized retail environments.

Technical Skills

Programming: Python, C++, Java, JavaScript, C#, Lua

Machine Learning: Computer Vision, Deep Learning, Natural Language Processing, Transformers

Frameworks: PyTorch, TensorFlow, Scikit-learn

Data Science: Data Analysis, Data Visualization, Model Evaluation

Tools: Linux, Git, Docker, Google Colab